



Series — DC

# 279kWh DC Battery Cabinet

Sodium-Ion NFPP technology engineered for critical infrastructure resiliency and demanding commercial deployments — with bankable economics.

**279kWh**

NAMEPLATE ENERGY

**229kWh**

USABLE (85% DOD)

**140kW**

MAX RATED POWER

**14%**

LOWER TCOO VS. LFP

**634–994V**

DC VOLTAGE RANGE

**2.2–10<sup>hr</sup>**

DURATION RANGE

**–40/+55°C**

OPERATING TEMP

**IP54**

ENCLOSURE RATING



ENERGY STORAGE YOU CAN BANK ON

[www.CoulombTechnology.com](http://www.CoulombTechnology.com)

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208-768-8888

UL 1973

UL 9540

UL 9540A

UN 38.3

IP54

IEC 62619

PRODUCT OVERVIEW

# 279kWh DC Battery Cabinet

- Sodium-Ion NFPP
- Air-Cooled
- 2.2–10 Hour Duration



Built for critical infrastructure resiliency and demanding use cases, the Series DC 279kWh Battery Cabinet prioritizes compliance, field-validated performance, and bankable warranties — delivering a **14% lifetime TCoO improvement** over liquid-cooled lithium-ion systems.

**Includes an outdoor, IP54-rated enclosure containing:**

- Sodium-Ion NFPP battery modules
- Air-cooled thermal management
- Fire suppression & gas detection
- Battery Management System (BMS)
- Auxiliary safety systems
- Explosion venting system

KEY BENEFITS



### Safety-First Design

Sodium-ion chemistry eliminates thermal runaway risk from the chemistry up — enabling safer siting and simpler AHJ approvals.

- ✓ Lower lifetime O&M costs
- ✓ Air-shippable — faster deployment
- ✓ Reduced thermal propagation risk
- ✓ Clearer AHJ approvals



### Extreme Performance

Operates where traditional batteries fail — from arctic cold to desert heat — without derating or HVAC infrastructure.

- ✓ -40°C to +55°C operation
- ✓ 20-year projected battery life
- ✓ No HVAC required
- ✓ Fast charging capability



### Supply Chain Resilience

Real-world validation, not laboratory promises. Clear domestic manufacturing pathway with no tariff exposure.

- ✓ DOD / Federal compliance path
- ✓ No tariff exposure
- ✓ Domestic manufacturing pathway
- ✓ Supply chain diversification

WHERE THE SERIES DC PERFORMS



MANUFACTURING / COLD STORAGE / DISTRIBUTION

#### Peak Demand Management

Reduce demand charges by 30–50% at manufacturing facilities, cold storage warehouses, distribution centers, and agriculture operations while maintaining full capacity in extreme temperatures.



DATA CENTERS / HEALTHCARE / TELECOM

#### Critical Infrastructure Backup

Guaranteed uptime for data centers, hospitals, emergency services, and telecommunications — with air-cooled simplicity and no fire suppression complexity.



FORKLIFT / WAREHOUSE / PORT / AIRPORT

#### Material Handling & Fleet

Fast charging and extreme temp tolerance for 24/7 operations — forklift fleets, warehouse automation, port equipment, and airport ground support.



SOLAR / WIND / CAMPUS / COMMUNITY

#### Renewable Energy Integration

Long-duration storage with 20-year design life for bankable projects — solar + storage microgrids, wind energy smoothing, campus energy systems, and community solar projects.

TECHNICAL DATA SHEET

Specifications



|                                                                                                                             |                                                                                      |                                                                                        |                                                                                            |
|-----------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------|
| <div>SERIES DC — CABINET</div> <div></div> | <div>NAMEPLATE ENERGY</div> <div>279kWh</div> <div>229 kWh usable at 85% DOD</div>   | <div>MAX RATED POWER</div> <div>140kW</div> <div>125 kW guaranteed constant</div>      | <div>DC VOLTAGE RANGE</div> <div>634-994v</div> <div>Flexible inverter compatibility</div> |
|                                                                                                                             | <div>DURATION RANGE</div> <div>2.2-10hr</div> <div>Peak shaving to long backup</div> | <div>OPERATING TEMPERATURE</div> <div>-40/+55°C</div> <div>No HVAC — any climate</div> | <div>ENCLOSURE RATING</div> <div>IP54</div> <div>Outdoor-rated direct install</div>        |

| PARAMETER                  | VALUE / DESCRIPTION                           | UNIT   |
|----------------------------|-----------------------------------------------|--------|
| NAMEPLATE ENERGY CAPACITY  | 279                                           | kWh    |
| USABLE ENERGY (85% DOD)    | 229                                           | kWh    |
| MAX RATED POWER            | 140                                           | kW     |
| GUARANTEED CONSTANT POWER  | 125                                           | kW     |
| DC VOLTAGE RANGE           | 634 – 994                                     | VDC    |
| DURATION                   | 2.2 – 10                                      | Hours  |
| CELL TYPE                  | Sodium-Ion NFPP (Non-Flammable Polymer Pouch) | —      |
| OPERATING TEMP (DISCHARGE) | -40°C to +55°C                                | —      |
| DIMENSIONS (L x W x H)     | 2.0 x 1.5 x 2.3                               | Meters |
| ENCLOSURE (IP RATING)      | Outdoor Cabinet — IP54                        | —      |
| THERMAL MANAGEMENT         | Air-Cooled (No HVAC Required)                 | —      |
| BMS COMMUNICATION          | US BMS, Modbus TCP                            | —      |
| AUXILIARY SYSTEMS          | Fire Suppression, Gas Detection, Venting      | —      |

|                                                                                                                                |                                                                                                        |                                                                                                                |
|--------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------|
| <div>CERTIFICATIONS</div> <div><div>UL 1973</div><div>UL 9540</div><div>UN 38.3</div></div> <div>Cell &amp; Module Level</div> | <div>UL 9540A TESTING</div> <div>Cell: Tested &amp; Passed</div> <div>Module/System: In Progress</div> | <div>WARRANTY</div> <div>Cell: 3 Years</div> <div>System: 5 Years</div> <div>Extended warranty available</div> |
|--------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------|

|                                                     |                                 |                                                |
|-----------------------------------------------------|---------------------------------|------------------------------------------------|
| FACTORY ACCEPTANCE TEST (FAT) SCOPE                 |                                 |                                                |
| ✓ Visual, Mechanical & Torque Inspection            | ✓ Protections & Isolation Tests | ✓ Functional & Conditioning Test               |
| ✓ 3 x 0.125C Capacity/Energy Verification (85% DOD) | ✓ Baseline OCV Measurement      | ✓ 10 x 0.5C Stability Cycles with Power Checks |
| Series DC — 279kWh DC Battery Cabinet   Rev. 2.0    |                                 |                                                |
| www.CoulombTechnology.com                           |                                 |                                                |
| sales@coulombtechnology.com   208-768-8888          |                                 |                                                |



TECHNOLOGY COMPARISON

Sodium-Ion NFPP vs. Lead-Acid

• Performance

• Safety

• Total Cost of Ownership



WHY SODIUM-ION NFPP OUTPERFORMS LEGACY LEAD-ACID TECHNOLOGY

A Clear Advantage Across Every Performance Dimension

Coulomb Technology's Sodium-Ion NFPP chemistry delivers superior cycle life, wider operating temperature range, zero maintenance, and a fundamentally safer chemistry — making it the ideal replacement for lead-acid in critical commercial and industrial applications.

| FEATURE              | SODIUM-ION NFPP         | LEAD-ACID                       |
|----------------------|-------------------------|---------------------------------|
| Cycle Life           | 3,000+ cycles @ 80% DoD | 300–500 cycles @ 50% DoD        |
| Weight (100Ah)       | 13 kg (28.7 lb)         | ~22 kg (48.5 lb)                |
| Usable Capacity      | 80–100% DoD             | 50% DoD (to preserve life)      |
| Discharge Temp Range | –40°C to +60°C          | –20°C to +50°C                  |
| Charge Temp Range    | –10°C to +45°C          | 0°C to +40°C                    |
| Maintenance          | Zero maintenance        | Regular watering & equalization |
| Self-Discharge       | <3% per month           | 5–15% per month                 |
| Charging Efficiency  | ~95%                    | ~75%                            |
| Thermal Runaway Risk | Zero risk               | Low risk                        |
| Toxic Gas Emissions  | None                    | Hydrogen gas during charging    |
| Memory Effect        | None                    | Sulfation if undercharged       |
| Lifespan             | 10+ years               | 3–5 years                       |
| Environmental Impact | Non-toxic, recyclable   | Lead & acid hazardous           |

10×

Longer cycle life  
vs. lead-acid

40%

Lighter weight  
per 100Ah

Zero

Maintenance  
required

95%

Round-trip  
efficiency

TOTAL COST OF OWNERSHIP

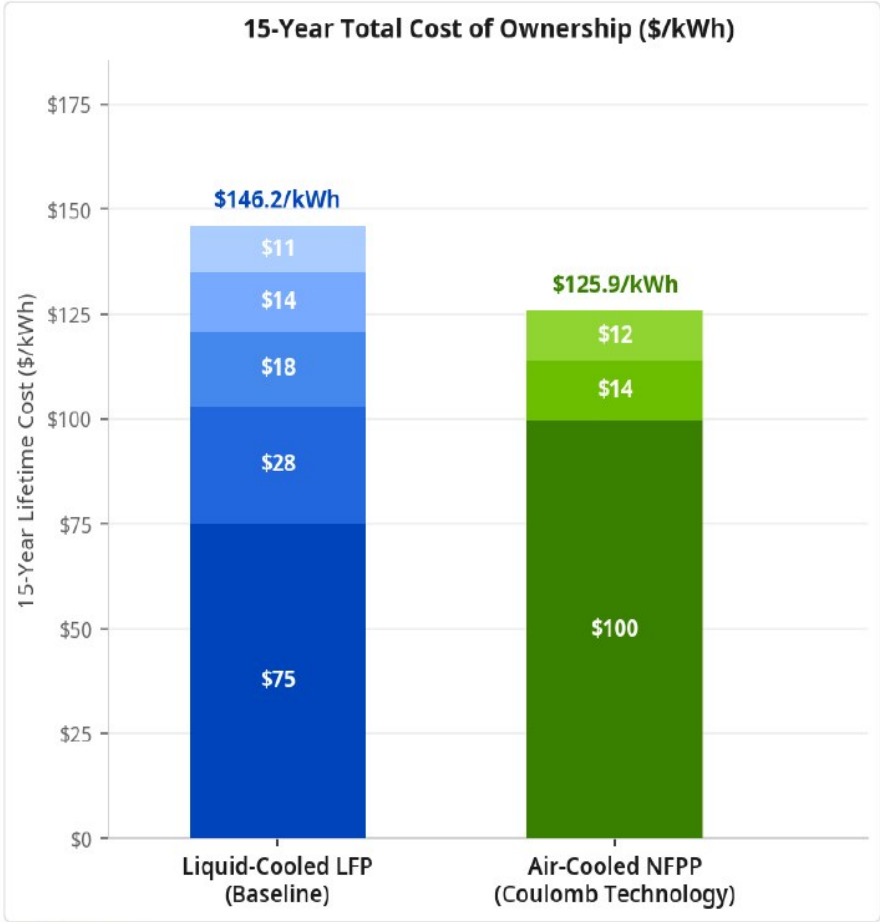
Bankable Economics



SYSTEM ADVANTAGE OVER LFP-BASED SYSTEMS

14% Lower Total Cost of Ownership Over the Project Lifetime

Our sodium-ion technology delivers measurable savings through eliminated HVAC requirements, reduced degradation, and lower maintenance, insurance, and permitting costs — validated by real-world deployments, not laboratory projections.



MODEL ASSUMPTIONS

|                            |          |
|----------------------------|----------|
| NFPP installed price delta | \$16/kWh |
| Cycles per year            | 365      |
| HVAC load delta            | 90%      |
| Insurance cost delta       | 20%      |
| Augmentation delta         | 20%      |
| O&M cost delta             | 30%      |
| Cell derating / cold temps | 75%      |
| Discount rate / NPV calc   | 8%       |

COST COMPONENTS

- LFP — CapEx
- LFP — HVAC Load
- LFP — O&M
- LFP — Insurance
- NFPP — CapEx
- NFPP — O&M
- NFPP — Insurance
- LFP — Augmentation

14% TCoO Savings — Air-cooled NFPP (\$125.9/kWh) vs. Liquid-cooled LFP (\$146.2/kWh) over a 15-year project lifetime.

\$20.3/kWh — Lifetime savings per kWh over project lifetime

20yr — Projected battery life, sodium-ion NFPP chemistry

RESEARCH & DEVELOPMENT PARTNERS

